

PROJECT BASED LEARNING

Review 1

A Secure Digital Platform for Maintaining Patient Treatment Records

GROUP MEMBERS



GROUP NO.

SY2-G__

UNDER THE GUIDANCE OF

Dr. / Ms. _____

Department of Computer Engineering

Field Visit: Deenanath Mangeshkar Hospital, Pune

Domain Introduction

H

Domain Overview

Healthcare Technology (Health IT) covers digital systems used to record, store, and exchange medical information. India is rapidly digitizing healthcare through initiatives such as the Ayushman Bharat Digital Mission (ABDM), pushing hospitals and clinics toward Electronic Health Records.

!

Problem Context

Despite this push, patient data remains fragmented across prescriptions, lab reports, and hospital files. This scattering directly affects diagnosis accuracy, treatment continuity, and patient safety - making it a pressing real-world concern.

S

Project Scope

This project focuses on patient-controlled digital health records - a secure platform where patients store treatment history and grant doctors permission-based access during consultations.

WHY THIS MATTERS

Problem Statement

“

Patients visit different doctors and healthcare facilities throughout their lives, resulting in medical information being scattered across prescriptions, diagnostic reports, hospital records, and personal documents.

Patients often forget previous treatments, medicines, allergies, diagnoses, or procedures - and when a doctor lacks access to this history, it increases the risk of inappropriate treatment or medication-related complications.

There is a clear need for a secure, centralized, patient-controlled digital record system.

RELEVANCE & IMPACT

- Improves patient safety by preventing missed medical history
- Reduces risk of incorrect treatment or drug interactions
- Saves doctors' time during consultations
- Supports India's shift toward digital healthcare (ABDM)
- Gives patients control over who accesses their data

WHAT THE PROJECT AIMS TO ACHIEVE

Objectives

1

Understand how patients currently maintain medical records and prescriptions

2

Identify difficulties faced by patients in recalling previous treatments

3

Understand how doctors obtain a patient's medical history during consultations

4

Study how records, prescriptions, and diagnoses can be stored digitally

5

Establish the importance of patient consent before sharing records

6

Identify the security and privacy requirements for protecting medical data

ACTIVITY 1

Field Visit & Problem Identification

LOCATION

Deenanath Mangeshkar
Hospital, Pune

DATE OF VISIT

TEAM SIZE

4 Members

The team visited Deenanath Mangeshkar Hospital to understand real-world challenges in maintaining and accessing patient treatment records. The visit involved direct interaction with hospital staff and observation of existing record-keeping practices, which helped ground the project's problem statement in a genuine clinical setting.

KEY TAKEAWAY

The visit confirmed that hospital staff and patients rely heavily on physical documents, and that missing history directly affects treatment decisions - validating the need for a secure, centralized, patient-controlled digital platform.

PHOTOS FROM THE VISIT

PHOTO

Geotagged Photo 1

PHOTO

Geotagged Photo 2

ACTIVITY 1 (CONTINUED)

Problems Identified During Visit

Incomplete Medical History

Patients forget past diagnoses, treatments or procedures over time

No Continuous Medical History

Records stay siloed with individual hospitals instead of following the patient

Scattered Medical Records

Prescriptions and reports stored separately as papers or in different systems

Lack of Patient-Controlled Access

No secure way for patients to decide who can view their data

Forgotten Prescriptions

Names/doses of past medicines are often not remembered accurately

Risk of Incorrect Treatment

Missing history can lead to incomplete treatment decisions

Difficulty Sharing Reports

Patients must carry physical reports to every new consultation

Privacy & Security Concerns

Sensitive data needs secure storage, authentication, and consent

CAN THIS PROJECT WORK?

Feasibility Study



Technical Feasibility

Cloud-hosted database with encryption, secure authentication (OTP/consent-based access), and a web/mobile front-end. Uses widely available technologies - manageable within team skillsets and timeline.



Commercial Feasibility

Low development cost using open-source frameworks; potential freemium model with hospitals/clinics as future partners. Aligns with rising demand for digital health solutions in India.



Operational Feasibility

Requires onboarding of patients and doctors, along with basic staff training. Minimal manpower needed for a pilot rollout; scalable to larger hospital networks later.



Scheduling Feasibility

Achievable in phases - requirement analysis, design, core development, testing, and pilot deployment - within the academic project timeline.

SWOT Analysis

S Strengths

- Centralized, always-accessible medical history
- Patient-controlled, consent-based access
- Reduces errors from missing treatment data
- Scalable cloud-based architecture

W Weaknesses

- Depends on internet connectivity
- Relies on hospitals digitizing existing records
- Migrating old paper records takes time
- Adoption curve for elderly / less tech-savvy users

O Opportunities

- Alignment with Ayushman Bharat Digital Mission
- Rising smartphone & internet penetration in India
- Partnerships with hospitals & diagnostic labs
- Future integration with insurance claim processing

T Threats

- Data privacy & security breach risks
- Regulatory compliance (DPDP Act, health data laws)
- Resistance from hospitals to share data externally
- Competition from established EHR vendors

RELATED WORK

Literature Survey

S.No.	Author(s) & Year	Title of Paper / Study	Methodology	Key Findings	Project Gap / Relevance
1	F. Masroor, A. Gopalakrishnan & N. Goveas, 2024	Building a Secure Electronic Health Record Management System on Cost-Effective Devices Using Non-Fungible Tokens	Blockchain-based NFTs provide authentication and controlled EHR access on low-cost devices; evaluated in a hospital scheduling scenario.	Empowers patients to control their EHRs while supporting secure access and authentication; evaluated access and scheduling delays.	Uses blockchain/NFT infrastructure and resource-constrained devices; our project targets a simpler web/mobile patient-record workflow.
2	S. Wu, A. Zhang, Y. Gao & X. Xie, 2024	Patient-centric medical service matching with fine-grained access control and dynamic user management	CP-ABE with policy hiding, secure attribute matching, malicious-doctor tracking and dynamic revocation.	Patients can define personalized access policies, use fine-grained access control, and withdraw untrustworthy doctors with low reported overhead.	Cryptographic design is technically complex; our project translates the patient-control concept into a simpler practical consent workflow.
3	Y. Dong, S. K. Mun & Y. Wang, 2023	A blockchain-enabled sharing platform for personal health records	Prototype PHR-sharing platform using blockchain, next-generation access control and patient-controlled security, privacy and consent granularity.	Demonstrates patient-managed consent and permission changes across records from multiple healthcare organizations; reports millisecond-range permission-change performance.	Blockchain scalability remains a limitation; our pilot avoids requiring a distributed blockchain network.
4	R. P. Puneeth & G. Parthasarathy, 2024	Blockchain-Based Framework for Privacy Preservation and Securing EHR with Patient-Centric Access Control	Encrypted EHR storage with IPFS, blockchain and smart-contract-based patient-centric access control.	Patients can manage and authorize access; framework combines confidentiality, integrity, availability and privacy-preserving sharing.	Adds blockchain/IPFS and cryptographic complexity; our project focuses on an easier-to-deploy patient-controlled platform.
5	U. N. C. Cobrado et al., 2024	Access control solutions in electronic health record systems: A systematic review	Systematic review of 20 EHR access-control studies using identification, authentication, authorization and accountability categories.	ABAC is the most common authorization approach; gaps include MFA, emergency access, patient consent, accountability and standards compliance.	Provides the broader research basis for our security requirements and highlights consent/accountability as areas our design should explicitly address.

Recent literature consistently emphasizes patient-controlled consent, fine-grained access control, secure authentication and auditable sharing. Blockchain and advanced cryptography provide strong security but introduce implementation complexity. This project therefore focuses on a simpler, patient-controlled treatment-record platform while retaining secure authentication, authorization and consent.

Scope

IN SCOPE

- Secure storage of treatment history, prescriptions, diagnostic reports & allergies
- Patient-controlled, permission-based access for doctors
- Doctor upload of new prescriptions and treatment details (with consent)
- Web / mobile platform usable by both patients and doctors
- Authentication and encrypted data storage

OUT OF SCOPE

- Integration with insurance claim processing (future phase)
- Offline / no-internet functionality
- AI-based diagnosis or treatment suggestions
- Nationwide hospital network integration in this phase
- Billing and payment processing modules

CONCLUSION

Bringing It All Together

Across the domain research, field visit, and feasibility & SWOT analysis, one theme was consistent: patients and doctors both struggle with fragmented, physically stored medical records - and this gap directly affects the quality of care.

1

Domain Study

Confirmed a genuine, government-backed push toward digital health records in India

2

Field Visit

Verified real-world pain points first-hand at Deenanath Mangeshkar Hospital

3

Feasibility & SWOT

Showed the project is technically achievable and strategically well-positioned

These activities have directly shaped the project direction toward a secure, patient-controlled digital treatment record platform.

References

1

Government of India, “Ayushman Bharat Digital Mission,” National Health Authority, 2021. Retrieved from <https://abdm.gov.in>

2

World Health Organization, “Global Strategy on Digital Health 2020-2025,” WHO, Geneva, 2021.

3

Ministry of Electronics and Information Technology, “Digital Personal Data Protection Act,” Government of India, 2023.

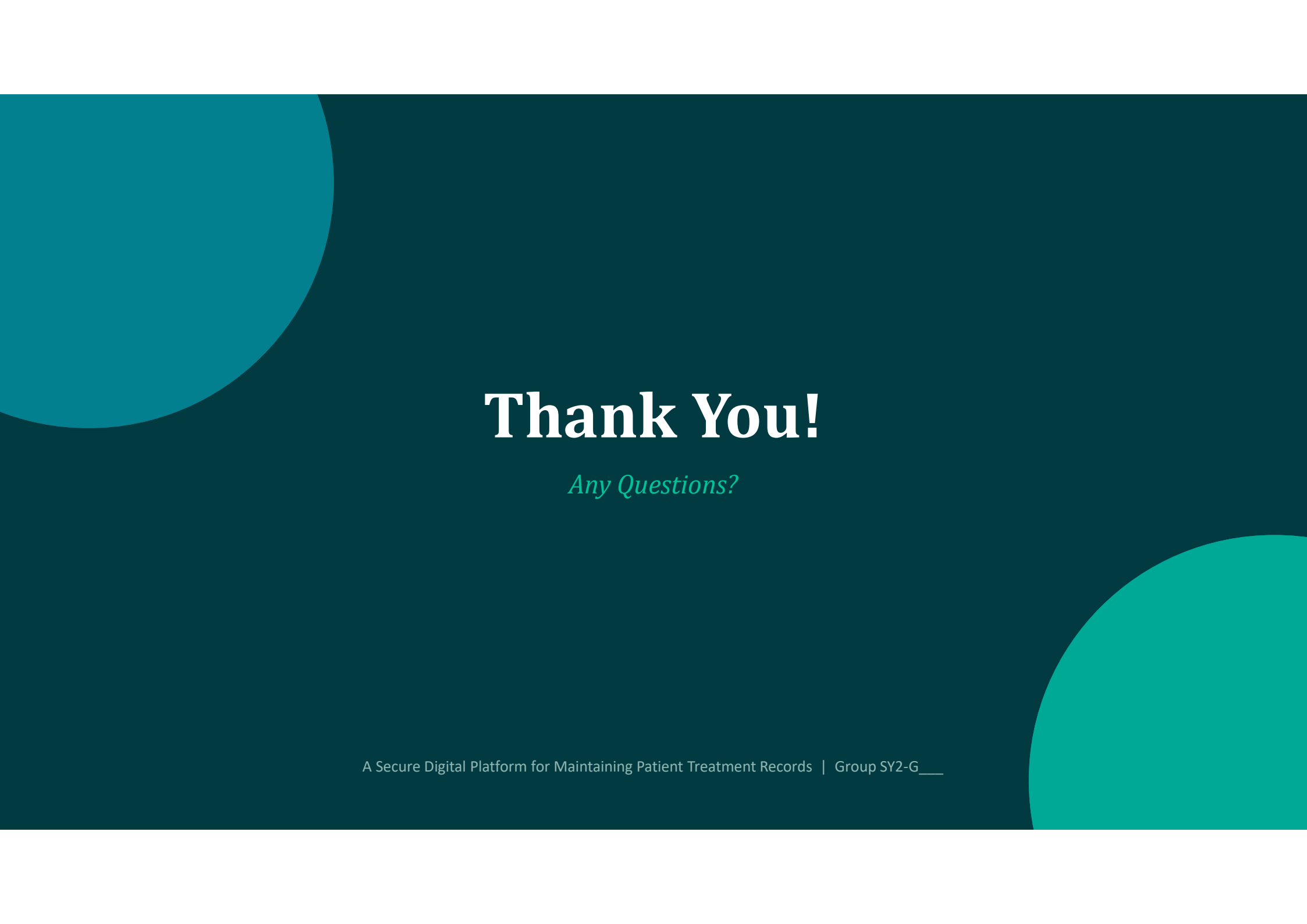
4

Field Visit to Deenanath Mangeshkar Hospital, Pune. (August 2026). Direct interaction with hospital staff.

5–

9

[5] F. Masroor, A. Gopalakrishnan, and N. Goveas, “Building a Secure Electronic Health Record Management System on Cost-Effective Devices Using Non-Fungible Tokens,” in Proc. 2024 IEEE Int. Conf. on E-health Networking, Application & Services (HealthCom), 2024, doi: 10.1109/HealthCom60970.2024.10880820.



Thank You!

Any Questions?

A Secure Digital Platform for Maintaining Patient Treatment Records | Group SY2-G____